THE UNIVERSITY OF QUEENSLAND
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This exam paper must not be removed from the venue

Venue _____
Seat Number _____
Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

School of Electrical Engineering & Computer Science
Semester 1 Examinations, 2025
COMP4702 / COMP7703 Machine Learning

This paper is for St Lucia Campus students.

Examination Duration: 180 minutes

Planning Time: 10 minutes

Exam Conditions:

•No written or printed material permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

1 x Gradescope Bubble Sheet

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Answer Part A in the Gradscope Bubble Sheet. Answer Part B in the space provided in the question paper.

For Examiner Use Only

Question	Mark

Total _____

Part A - Multiple choice questions (2 marks each). Record your answers on the bubble sheet provided.

Question 1.

The CIFAR-100 dataset consists of images from 100 different labeled classes, with 50000 images for training and 10000 for testing. Each image is 32×32 with Red, Green and Blue intensity values for each pixel. Assume that each image is "flattened" into a vector to create the input for a data point. If we denote a training set by $\mathcal{T} = \{\mathbf{x}_i, y_i\}_{i=1}^n$, $\mathbf{x} = [x_1, \ldots, x_p]^\top$, then for CIFAR-100:

(a) $n = 100, p = 32$.

(b) $n = 50000, p = 3072$.

(c) $n = 10000, p = 1024$.

(d) $n = 50000, p = 2048$.

Question 2.

The k-Means clustering algorithm can be seen as a simplified case of which other algorithm that we have studied during the course:

(a) Stochastic gradient descent.

(b) Principal component analysis.

(c) Recursive binary splitting.

(d) The Expectation Maximisation algorithm.

Question 3.

Let $g(\mathbf{x}_i; \boldsymbol{\theta})$ be the output of a standard logistic regression model. At the decision boundary, the output of this model is equal to:

(a) 0.5.

(b) 0.

(c) 1.

(d) -1.

Question 4.

Which of the following is *not* an example of a hyperparameter?

(a) The value of k in k-nearest neighbours.

(b) The number of layers in a feed-forward neural network.

(c) The offset term in linear regression, θ_0.

(d) The filter size in a convolutional neural network layer.

Question 5.

Which of the following is a commonly-used measure of the performance of a binary classifier:

(a) The exponential loss function.

(b) Area under the ROC curve.

(c) The magnitude of the gradient with respect to the model parameters after training.

(d) The F_1 score divided by the mean-squared error.

Question 6.

A commonly used loss function in machine learning is given by:

$$L(y, \hat{y}) = \mathbb{I}\{\hat{y} \neq y\} = \begin{cases} 0 & \text{if } \hat{y} = y. \\ 1 & \text{if } \hat{y} \neq y. \end{cases}$$

This is known as the:

(a) Misclassification loss.

(b) Mean squared error.

(c) ϵ-insensitive loss.

(d) Cross-entropy loss.

Question 7.

Random forests aim to improve performance over simple bagged ensembles by:

(a) Allowing the decision tree base models to grow to a larger depth.

(b) Repeatedly reshuffling the data prior to training each base model.

(c) Implementing aggressive pruning within the base models, so that the forest can be seen for the trees.

(d) Reducing the correlation between the base models in the ensemble.

Question 8.

Gaussian Processes differ from most machine learning models in that they:

(a) Use maximum likelihood estimation to formulate the training problem.

(b) Have no hyperparameters.

(c) Produce a distribution over the function space for the output variable.

(d) Can be trained via stochastic gradient descent.

Question 9.

Principal component analysis (PCA) can be performed by calculating the eigenvectors and eigenvalues of the covariance matrix of a data set. The eigenvalues measure:

(a) The mean squared error of the model on the data.

(b) The fraction of variance captured by the respective principal components.

(c) The amount of memory required to store the data after performing dimensionality reduction.

(d) The angle of rotation of each principal component compared to the original coordinate system.

Question 10.

Regarding auto-encoders, which of the following statements is *false*?

(a) An auto-encoder is a type of feedforward neural network.

(b) Auto-encoders can be trained via gradient descent.

(c) Auto-encoders can be used to perform dimensionality reduction.

(d) The weights of an autoencoder must be initialized using a Gaussian distribution prior to training.

(next question over)

Part B - Questions carry the number of marks indicated. Write your answers in the space provided after each part of each question.

Question 1.

Figure 1 (from a book by Kevin Murphy) shows some results of performing k-nearest neighbor (k-NN) classification on a 2-D dataset.

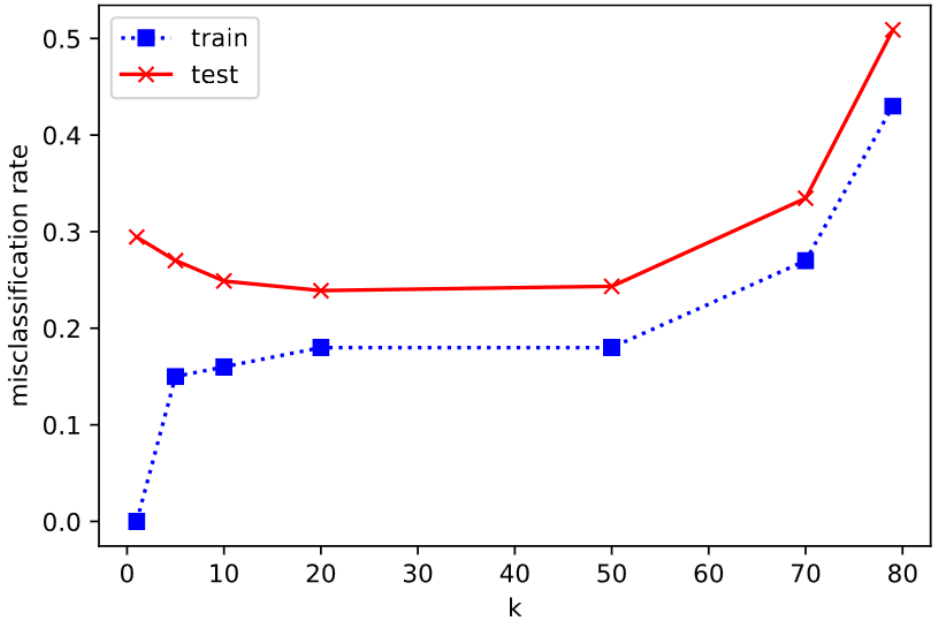


Figure 1

(a) Explain the shape of the test curve shown. (4 marks)

(question continues over)

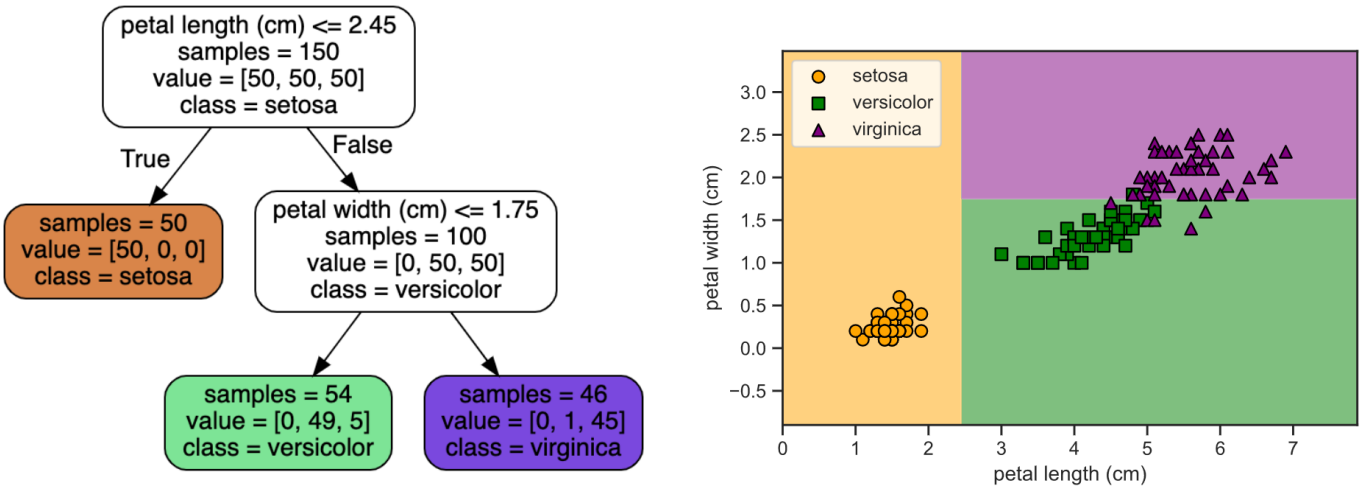
(b) The training curve in Figure 1 has a misclassification rate of 0 at $k = 1$. Explain how this is possible and hence would be true for *any* dataset. (4 marks)

(c) Figure 1 shows that as k increases, the misclassification rate on the training set increases. What does this suggest about the relationship between k, model complexity overfitting and underfitting? (4 marks)

(next question over)

Question 2.

Figure 2 (from a book by Kevin Murphy) shows a decision tree trained on the Iris dataset (a), together with the data and decision regions (b) produced by the tree with respect to two of the input features (petal width and length). Note that leaf nodes show the number of data points (samples) classified by each node, the number of points classified into each class ([setosa, versicolor, virginica]) and the classification produced.



(a) *(b)*

Figure 2

(a) Given a test point $\mathbf{x}' = [\text{petal length} = 3\text{cm}, \text{petal width} = 2.5\text{cm}]$, what is the predicted output from the decision tree and explain how it arrives at this output. (5 marks)

(question continues over)

(b) What is the misclassification rate of the tree? (3 marks)

(c) Draw the confusion matrix for this data, including labels for each row and column. (5 marks)

(next question over)

Question 3.

Given a labeled dataset, a Gaussian mixture model can be trained to perform classification by fitting the parameters of the model using maximum likelihood estimates. This is a form of supervised learning, since the class labels are used. Assume that an additional set of unlabelled data becomes available, so that we now have:

$$\mathcal{T} = \{\{\mathbf{x}_i, y_i\}_{i=1}^{n_l}, \{\mathbf{x}_i\}_{i=n_l+1}^{n}\}$$

Explain how learning in the Gaussian mixture model can be modified to perform semi-supervised learning using all of the data available, $\mathcal{T}$. You should use the notation given to assist with your explanation. You may also use pseudocode or a diagram but this is not necessary. (6 marks)

(next question over)

Question 4.

Figure 3 shows linear regression applied to a 1-D dataset. Using the notation from lectures, the cost function for training this model can be written as:

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^{n} L(\hat{y}(\mathbf{x}_i; \boldsymbol{\theta}), y_i)$$

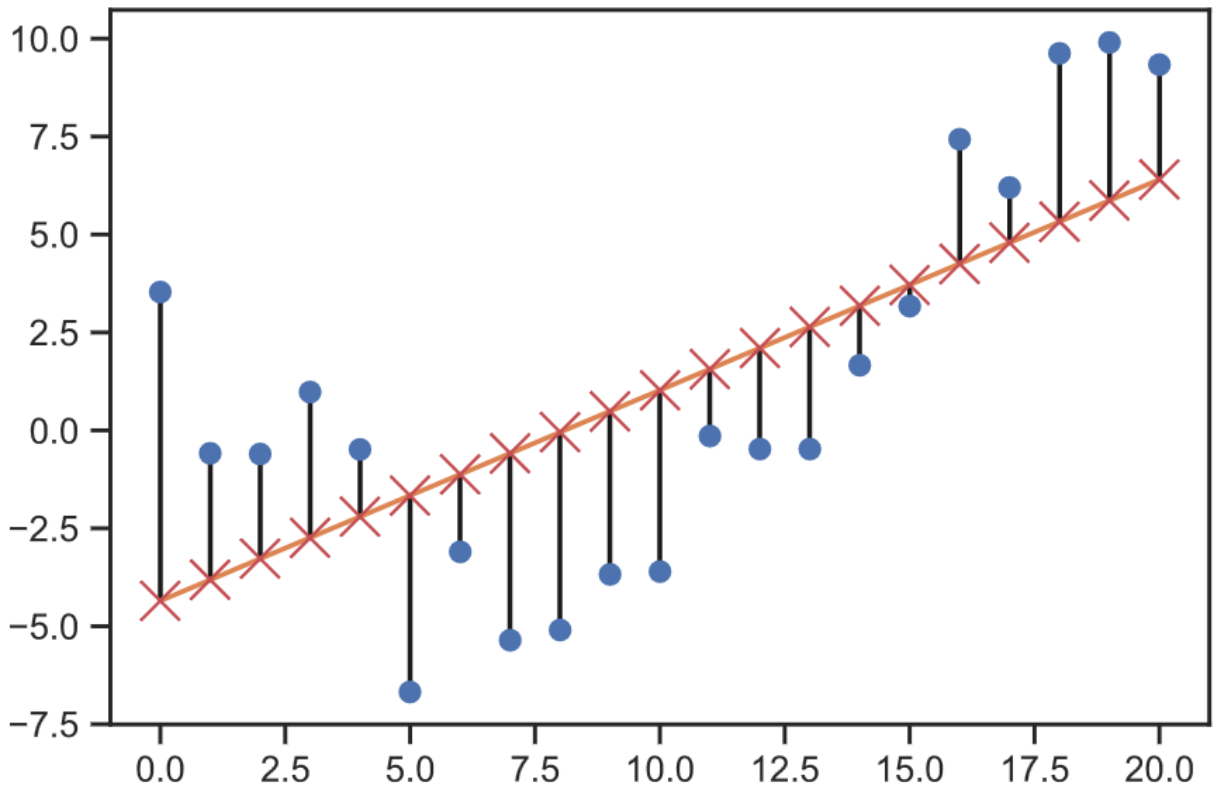


Figure 3

(a) Label the axes in Figure 3. (2 marks)

(b) Draw arrows and annotate Figure 3 to show an examples of (i) $\mathbf{x_i}$; (ii) y_i; (iii) $\hat{y}(\mathbf{x}_i; \boldsymbol{\theta})$; (iv) $\hat{y}(\mathbf{x}_i; \boldsymbol{\theta}) - y_i$. (4 marks)

(c) State the value of n from Figure 3. (2 marks)

(next question over)

Question 5.

Figure 4 (a,b,c) shows different polynomial regression models trained on a 1-D dataset. The Mean Squared Error versus the degree of polynomial for a range of models is also shown (d).

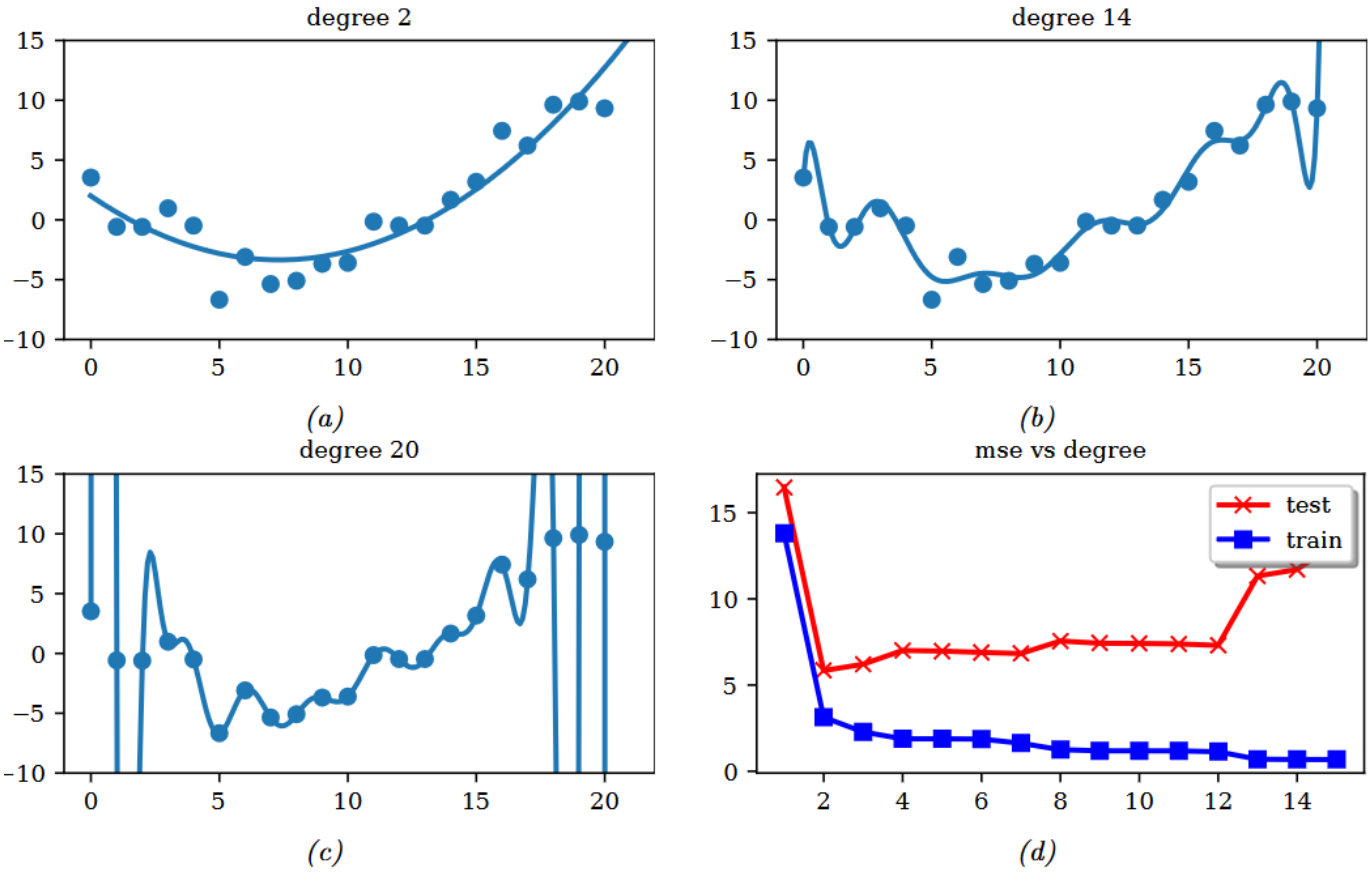


Figure 4

(a) Use Figure 4 to make an argument as to why the training set error does NOT provide a good estimate of the generalisation error of a machine learning model? (4 marks)

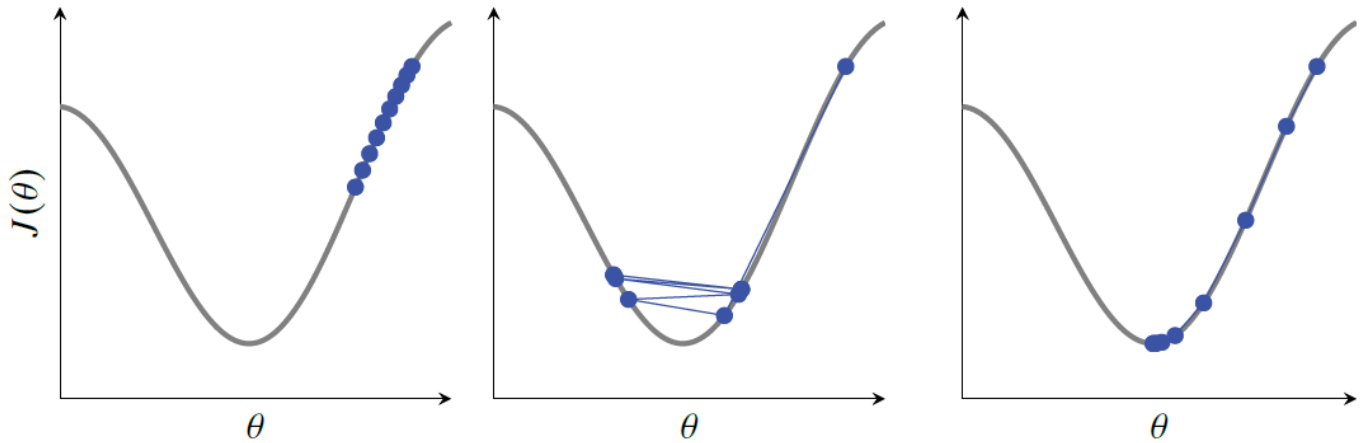
(question continues over)

(b) Use Figure 4 to give a short explanation for the phenomena of underfitting, overfitting and model complexity in machine learning. (4 marks)

(next question over)

Question 6.

Figure 5 (from the textbook) shows an illustration of gradient descent when different learning rate values are used.



(a) Low learning rate $\gamma = 0.05$ **(b)** High learning rate $\gamma = 1.2$ **(c)** Good learning rate $\gamma = 0.3$

Figure 5

Using the example of training a feedforward neural network and Figure 5, explain briefly what the objective of gradient descent is and why it can be challenging to find a suitable value for the learning rate hyperparameter. (5 marks)

(next question over)

Question 7.

Figure 6 (from the textbook) shows a diagram of a feed-forward neural network with a single hidden layer.

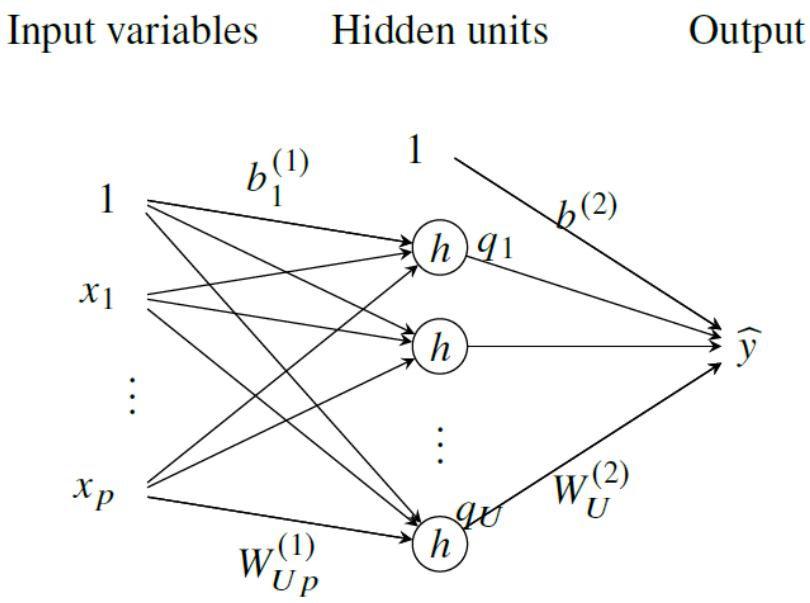


Figure 6

Using the notation given in Figure 6, write down an equation for the output of the neural network. (4 marks)

(question continues over)

Question 8.

Back in the 1980's, 8-bit computers commonly used an 8×8 bitmap (pixels) to represent ASCII characters (including a total of 95 different printable characters).

(a) Consider building a feed-forward neural network that uses this data as input and tries to classify each input as one of the printable characters. You decide to use two hidden layers with 50 units in each layer. Calculate the total number of weights in the network, including bias weights. Show your working. (3 marks)

(b) Now consider using a convolutional neural network for this data. You decide that the network architecture will have:

- The input layer.
- Convolutional layer 1, that has 6 filters with a filter size of 5×5, zero padding of size 2 all around the bitmap and a stride of 2.
- Convolutional layer 2, that has 4 filters with a filter size of 3×3, no padding and a stride of 1.
- A fully-connected layer of 10 hidden units.
- The output layer that uses softmax activation to produce probabilistic outputs.

Calculate the total number of weights in the network, including bias weights. Show your working. (5 marks)

(next question over)

Question 9.

Many ensemble methods in machine learning use the idea of bootstrapping. Explain briefly why this is an effective way of reducing model error in terms of the bias-variance decomposition. (4 marks)

Question 10.

The notion of a kernel plays an important role in a number of machine learning methods, including support vector machines. Give a short, intuitive explanation for the meaning of a kernel in machine learning: what does a kernel allow us to do given some data (perhaps non-numerical) from a given problem? (4 marks)

(next question over)

Question 11.

Figure 7 shows a summary of Bayesian linear regression from the textbook. Use this to help you explain the main difference between "standard" linear regression (using maximum likelihood or least squares training) and Bayesian linear regression. (4 marks)

Learn Bayesian linear regression

Data: Training data $\mathcal{T} = \{\mathbf{x}_i, y_i\}_{i=1}^n$
Result: Posterior $p(\boldsymbol{\theta} \mid \mathbf{y}) = \mathcal{N}(\boldsymbol{\theta}; \boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n)$

1 Compute $\boldsymbol{\mu}_n$ and $\boldsymbol{\Sigma}_n$ as (9.10).

Predict with Bayesian linear regression

Data: Posterior $p(\boldsymbol{\theta} \mid \mathbf{y}) = \mathcal{N}(\boldsymbol{\theta}; \boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n)$ and test input $\mathbf{x}_\star$
Result: Posterior predictive $p(f(\mathbf{x}_\star) \mid \mathbf{y}) = \mathcal{N}(f(\mathbf{x}_\star); m_\star, s_\star)$

1 Compute $m_\star$ and $s_\star$ as (9.11).

Method 9.1: Bayesian linear regression.

Figure 7

Question 12.

Generative Adversarial Networks (GANs) typically involve two deep neural networks: the critic and the generator. Explain briefly what the role of the critic is in the GAN training process. (4 marks)

END OF EXAMINATION

This page can be used if you need extra space to answer any of the questions. Clearly indicate the question(s) that you are answering. Ensure that you also put a note within the body of the relevant question(s) directing the marker here.